

Inventory Management System

Schema Diagram

User
user_id/ email (PK)
password
name
role
hostel_id

Hostel
hostel_no. (PK)
hostel_name

Inventory_Item
item_id (PK)
name
category (FK)
quantity
last_updated

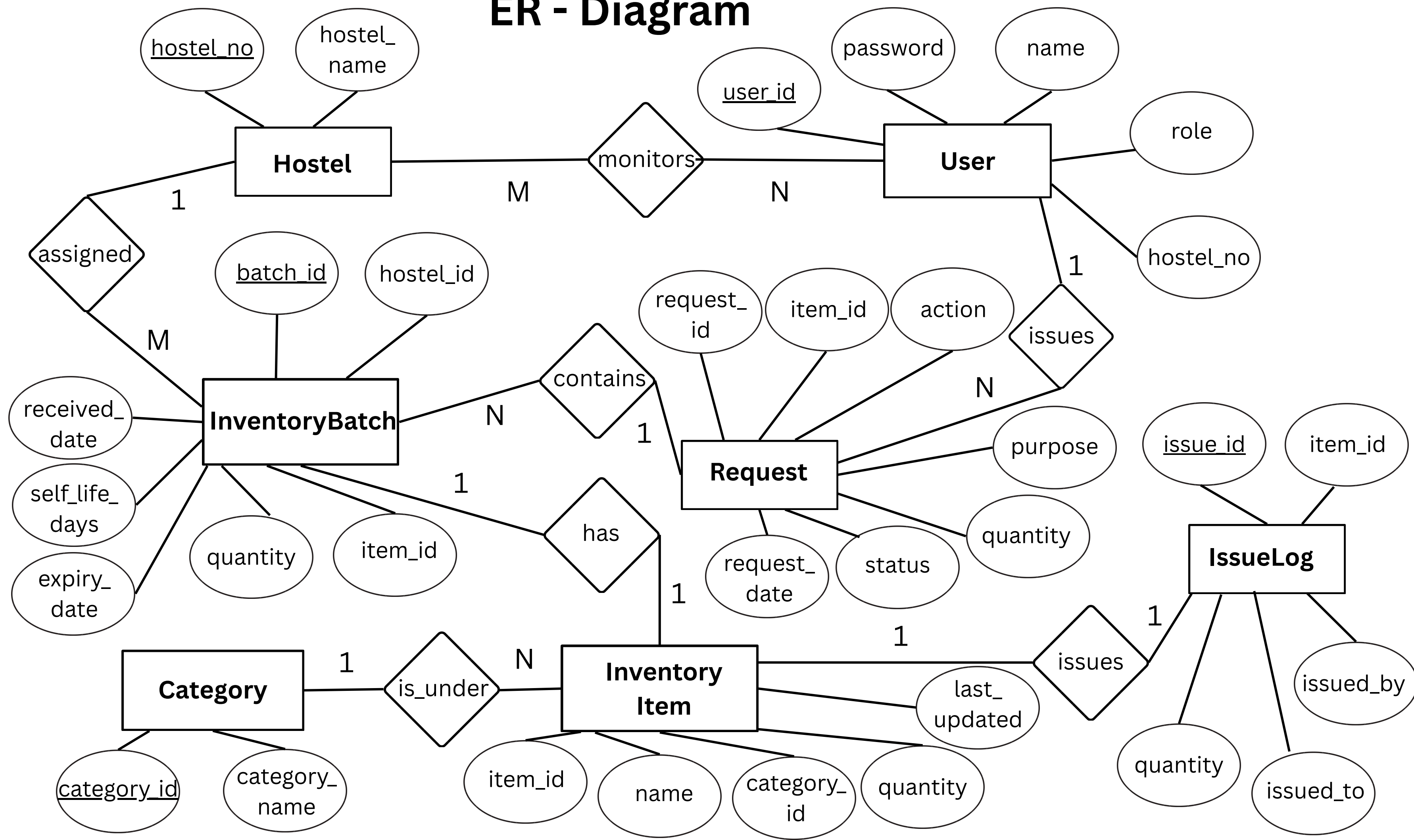
Category
category_id (PK)
category_name

Request
request_id (PK)
item_id (FK)
action
quantity
purpose
status
request_date
decision_date
decision_by (FK)

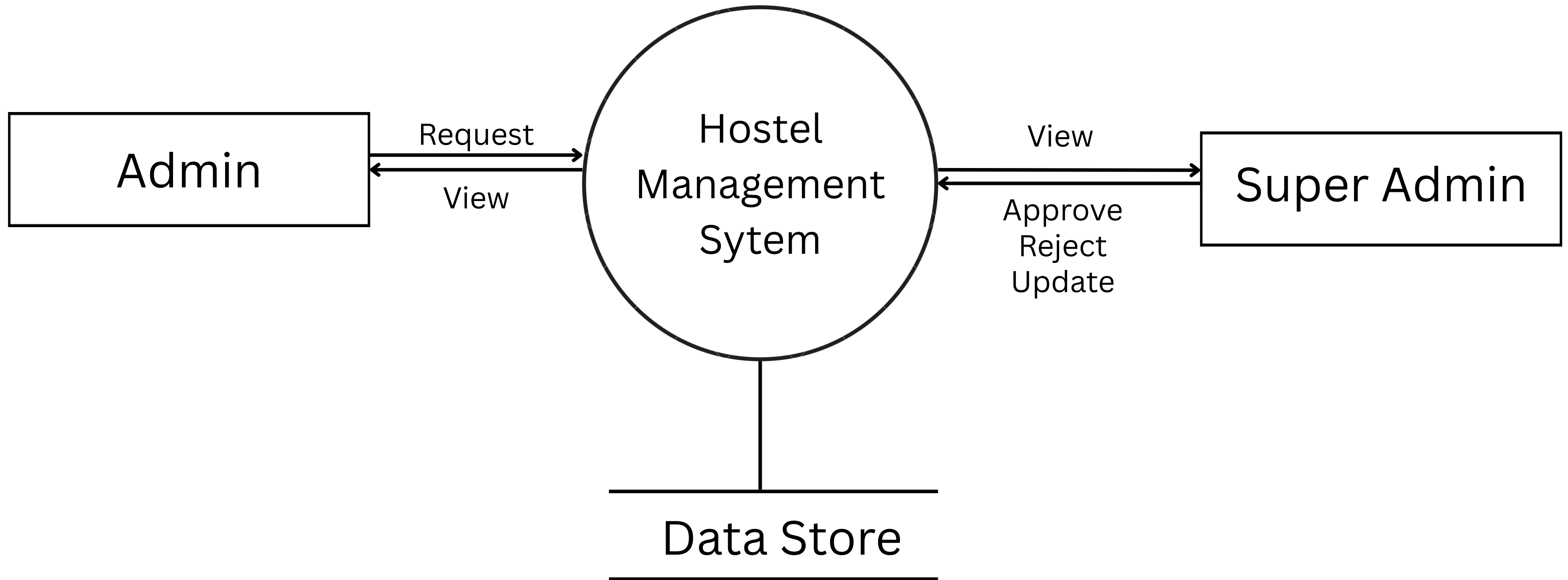
Inventory_Batch
batch_id (PK)
item_id (FK)
hostel_id (FK)
quantity
received_date
self_life_days
expiry_date

IssueLog
issue_id (PK)
item_id (FK)
hostel_id (FK)
quantity
issued_by
issued_to
reason
issue_date
created_at

ER - Diagram



Level-0 DFD



Level - 1 DFD (in progress)

API Endpoints

Item
GET /api/items – View all item types (with filters like category)
POST /api/items – Add a new item type
GET /api/items/:id – View item details
PUT /api/items/:id – Update item name/category/unit
DELETE /api/items/:id – Delete item type

Expiry/Low Stock Handling
GET /api/items/expired – View all expired batches
GET /api/items/low-stock – View items below min threshold

Inventory Batch
GET /api/batches – View all item batches
POST /api/batches – Add new stock (includes received date, quantity, shelf life)
GET /api/batches/:id – View batch details
PUT /api/batches/:id – Update quantity (e.g., after partial usage)
DELETE /api/batches/:id – Delete or archive expired/corrected batch

Item Request
POST /api/requests – Submit item request (admin)
GET /api/requests – View all requests (admin & super admin, filterable)
GET /api/requests/:id – View request details
PUT /api/requests/:id/approve – Super Admin approves request
PUT /api/requests/:id/reject – Super Admin rejects request

Frontend Pages

Inventory Dashboard Page

/inventory

- View item types
- Filter by hostel/category
- See low stock or expired badges

Add Inventory Item Page

/inventory/add

- Add new item type
(name, category, unit)

Inventory Batches Page

/inventory/:itemId/batches

- List batches of selected item
- Show expiry status, received date

Add Inventory Batch Page

/inventory/:itemId/add-batch

- Add a new batch
(quantity, received date, shelf life)

Request List Page

/requests

- Admin: View submitted requests
- Super Admin: View and filter by status

Submit Request Page

/requests/new

- Admin: Select item, quantity, purpose

Request Details Page

/requests/:id

- Show full request info, decision, history

Approve/Reject Request Page

/requests/:id/decision

- Approve or reject a pending request
- Optional: add remarks

Expired Items Page

/inventory/expired

- Highlight all expired batches

Expired Items Page

/inventory/expired

- Highlight all expired batches